

Similar Triangles — Solutions

Full Step-by-Step | Trap Alerts | Key Insights | Fast Methods | Desmos Verdicts

Q1

Easy

Answer: B) 15

Step-by-Step Solution:

Since triangles JKL and MNP are similar, corresponding sides are proportional.

$$JK/MN = KL/NP$$

$$6/10 = 9/NP$$

$$6 \times NP = 10 \times 9 = 90$$

$$NP = 90/6 = 15$$

Why Other Options Are Wrong:

A) 12 — incorrectly adds 4 to each side (additive, not multiplicative).

C) 18 — multiplies 9 by 2 instead of using the correct ratio.

D) 20 — confuses NP with MN scaled differently.

TRAP ALERT

Students often add the difference between corresponding sides instead of multiplying by the scale factor. Similarity is multiplicative, never additive.

KEY INSIGHT

Set up a proportion: $\text{side1}/\text{side2} = \text{side3}/\text{side4}$. Cross-multiply and solve.

FAST METHOD

Scale factor = $10/6 = 5/3$. Multiply KL by $5/3$: $9 \times 5/3 = 15$.

VERDICT

Hand-solve. Simple cross-multiplication takes 15 seconds. Desmos is overkill here.

Q2

Easy

Answer: C) 20

Step-by-Step Solution:

Scale factor = $3 : 5$ means larger/smaller = $5/3$.

Shortest side of larger = $12 \times (5/3) = 60/3 = 20$.

Why Other Options Are Wrong:

A) 7.2 — divides 12 by $5/3$ (inverts the ratio).

B) 15 — adds 3 to 12 (additive error).

D) 36 — multiplies 12 by 3 instead of $5/3$.

TRAP ALERT

The ratio 3:5 means the SMALLER triangle has sides in proportion 3 and the LARGER has 5. Students invert this and divide by $5/3$ instead of multiplying.

KEY INSIGHT

Identify which triangle is smaller and which is larger from the ratio before calculating.

FAST METHOD

Multiply the known side by the ratio of larger to smaller: $12 \times 5/3 = 20$.

VERDICT

Hand-solve. Direct ratio multiplication — 10 seconds.

Q3

Easy

Answer: C) 12.5

Step-by-Step Solution:

Since $DE \parallel BC$, triangles ADE and ABC are similar (AA similarity — corresponding angles are equal).

$$AD/AB = DE/BC$$

$$AB = AD + DB = 4 + 6 = 10$$

$$4/10 = 5/BC$$

$$4 \times BC = 50$$

$$BC = 50/4 = 12.5$$

Why Other Options Are Wrong:

A) 7.5 — uses $AD/DB = 4/6$ and multiplies $5 \times 6/4 = 7.5$ (uses wrong segment in ratio).

B) 8 — adds 3 to 5.

D) 15 — uses $DB/AD = 6/4$ and calculates $5 \times 3 = 15$ (uses ratio of part to part instead of part to whole).

TRAP ALERT

The ratio must be AD/AB (part to whole), not AD/DB (part to part). Many students use the wrong denominator.

KEY INSIGHT

When a line is parallel to the base and cuts two sides, use the Triangle Proportionality Theorem: the ratio of the segment to the WHOLE side equals the ratio of the parallel segments.

FAST METHOD

Scale factor = $AB/AD = 10/4 = 2.5$. So $BC = DE \times 2.5 = 5 \times 2.5 = 12.5$.

VERDICT

Hand-solve. One proportion setup — 20 seconds.

Q4

Easy

Answer: D) 21

Step-by-Step Solution:

R corresponds to U, S to V, so RS corresponds to UV and ST corresponds to VW.

$$RS/UV = ST/VW$$

$$8/12 = 14/VW$$

$$8 \times VW = 12 \times 14 = 168$$

$$VW = 168/8 = 21$$

Why Other Options Are Wrong:

- A) 16 — adds 2 to 14.
- B) 18 — uses wrong scale factor.
- C) 20 — adds 6 to 14 (additive approach).

TRAP ALERT

Students sometimes pair the wrong sides. Always identify which vertex corresponds to which before setting up proportions.

KEY INSIGHT

The problem tells you the correspondence: R to U, S to V. This means $RS \rightarrow UV$ and $ST \rightarrow VW$.

FAST METHOD

Scale factor = $12/8 = 3/2$. $VW = 14 \times 3/2 = 21$.

VERDICT

Hand-solve. Simple scale factor multiplication — 15 seconds.

Q5

Easy

Answer: 12

Step-by-Step Solution:

AB corresponds to DE, so BC corresponds to EF.

$$AB/DE = BC/EF$$

$$15/9 = 20/EF$$

$$15 \times EF = 9 \times 20 = 180$$

$$EF = 180/15 = 12$$

TRAP ALERT

Grid-in trap: students may compute the scale factor as $9/15$ and forget to apply it to BC. Always double-check which side you are solving for.

KEY INSIGHT

Scale factor = $DE/AB = 9/15 = 3/5$. Apply to BC: $20 \times 3/5 = 12$.

FAST METHOD

Quick mental math: $20 \times 3/5 = 12$.

VERDICT

Hand-solve. Ratio arithmetic — 15 seconds. No Desmos needed.

Q6

Medium

Answer: B) 6

Step-by-Step Solution:

When an altitude is drawn from the right angle to the hypotenuse, it creates two triangles similar to each other and to the original.

By the geometric mean relationship: $QS^2 = PS \times SR$

$$QS^2 = 4 \times 9 = 36$$

$$QS = \sqrt{36} = 6$$

Why Other Options Are Wrong:

- A) 5 — incorrectly averages PS and SR: $(4 + 9)/2 = 6.5$ then rounds.
- C) 6.5 — uses arithmetic mean $(4 + 9)/2$ instead of geometric mean.
- D) 13 — adds PS + SR to get hypotenuse length, not the altitude.

TRAP ALERT

The altitude to the hypotenuse gives a GEOMETRIC mean (sqrt of the product), not an ARITHMETIC mean (average). This is the #1 trap on altitude-to-hypotenuse questions.

KEY INSIGHT

Altitude to hypotenuse rule: $\text{altitude}^2 = (\text{segment 1})(\text{segment 2})$. Memorize this shortcut.

FAST METHOD

$$QS = \sqrt{4 \times 9} = \sqrt{36} = 6. \text{ One calculation.}$$

VERDICT

Hand-solve. This is a memorized formula — 10 seconds.

Q7

Medium

Answer: B) 15

Step-by-Step Solution:

The sun creates parallel rays, so the two objects and their shadows form similar triangles.

$$\text{height of flagpole} / \text{shadow of flagpole} = \text{height of post} / \text{shadow of post}$$

$$h/18 = 5/6$$

$$6h = 90$$

$$h = 15 \text{ feet}$$

Why Other Options Are Wrong:

- A) 12 — multiplies $5 \times 18/6$ but makes arithmetic error.
- C) 21.6 — uses $18/5 \times 6 = 21.6$ (inverts the proportion).
- D) 54 — multiplies 18×3 (3 from $18/6$) instead of 5×3 .

TRAP ALERT

Students sometimes set up the proportion upside down (shadow/height instead of height/shadow). Either way works IF consistent, but mixing them gives wrong answers.

KEY INSIGHT

Shadow problems always involve similar triangles because sunlight rays are parallel. Set up: object height / shadow length = constant for all objects.

FAST METHOD

$$\text{Ratio} = 5/6. \text{ Flagpole height} = 18 \times 5/6 = 15.$$

VERDICT

Hand-solve. Classic shadow problem — 15 seconds.

Q8

Medium

Answer: D) 15**Step-by-Step Solution:**

First find AC. A(0,0) and C(0,6), so $AC = 6$.

Scale factor = $5/2$.

Corresponding side in DEF = $6 \times 5/2 = 15$.

Why Other Options Are Wrong:

A) 2.4 — divides 6 by $5/2$ instead of multiplying.

B) 10 — applies scale factor to AB instead of AC.

C) 12 — uses the wrong scale factor or wrong side.

TRAP ALERT

Read carefully which side corresponds to which. AC is the vertical side (length 6), not the hypotenuse.

KEY INSIGHT

With coordinates, calculate the side length first, then apply the scale factor.

FAST METHOD

$AC = 6$ (vertical distance). $6 \times 5/2 = 15$.

VERDICT

Hand-solve. Find distance + multiply — 15 seconds.

Q9

Medium

Answer: C) 22.5**Step-by-Step Solution:**

Scale factor = $54/36 = 3/2$.

Longest side of larger triangle = $15 \times 3/2 = 22.5$.

Why Other Options Are Wrong:

A) 10 — divides 15 by $3/2$ (inverts ratio).

B) 18 — adds 3 to 15.

D) 25 — uses incorrect scale factor.

TRAP ALERT

The ratio of perimeters equals the ratio of corresponding sides (not areas). Some students square the ratio, which is wrong here.

KEY INSIGHT

Perimeter ratio = side ratio for similar figures. Area ratio = $(\text{side ratio})^2$.

FAST METHOD

$54/36 = 3/2$. Answer = $15 \times 3/2 = 22.5$.

VERDICT

Hand-solve. Direct proportion — 10 seconds.

Q10

Medium

Answer: 6**Step-by-Step Solution:**

$FG \parallel HI$ means triangles EFG and EHI are similar.

$$EF/EH = EG/EI$$

$$EH = EF + FH = 7 + 3 = 10$$

$$7/10 = 14/EI$$

$$EI = 14 \times 10/7 = 20$$

$$GI = EI - EG = 20 - 14 = 6$$

TRAP ALERT

The question asks for GI , not EI . After finding $EI = 20$, you must subtract $EG = 14$ to get $GI = 6$. Many students stop at 20.

KEY INSIGHT

Always re-read what the question asks for. Finding the whole side and then subtracting is a common extra step.

FAST METHOD

Scale factor = $10/7$. $EI = 14 \times 10/7 = 20$. $GI = 20 - 14 = 6$.

VERDICT

Hand-solve. Proportion + subtraction — 20 seconds.

Q11

Medium

Answer: C) 24**Step-by-Step Solution:**

Triangles WXY and WZA are similar, sharing angle W .

$$WX/WZ = WY/WA$$

$$10/15 = 16/WA$$

$$10 \times WA = 15 \times 16 = 240$$

$$WA = 240/10 = 24$$

Why Other Options Are Wrong:

A) 18 — adds 2 to 16.

B) 21 — adds 5 to 16.

D) 25 — multiplies 16 by $15/10$ then rounds incorrectly.

TRAP ALERT

Make sure to match corresponding vertices correctly. W is the shared vertex, so WX corresponds to WZ and WY corresponds to WA .

KEY INSIGHT

When two triangles share a vertex, the sides emanating from that vertex are the ones you pair in the proportion.

FAST METHOD

Scale = $15/10 = 3/2$. WA = $16 \times 3/2 = 24$.

VERDICT

Hand-solve. Simple ratio — 15 seconds.

Q12

Medium

Answer: C) 147

Step-by-Step Solution:

The ratio of areas of similar figures = (ratio of sides)².

Side ratio = 4:7, so area ratio = 16:49.

$48/\text{Area} = 16/49$

Area = $48 \times 49/16 = 48 \times 3.0625 = 147$

Why Other Options Are Wrong:

A) 84 — multiplies 48 by $7/4$ (uses side ratio instead of area ratio).

B) 112 — uses an incorrect calculation.

D) 196 — squares 14 (confused with a different quantity).

TRAP ALERT

Area scales by the SQUARE of the side ratio, not the side ratio itself. This is the most common similar-figures trap.

KEY INSIGHT

Memorize: sides scale by k , perimeters by k , areas by k^2 , volumes by k^3 .

FAST METHOD

Area ratio = $(7/4)^2 = 49/16$. Larger area = $48 \times 49/16 = 3 \times 49 = 147$.

VERDICT

Hand-solve. Square the ratio and multiply — 20 seconds. Desmos can help if you distrust your arithmetic.

Q13

Medium

Answer: B) 10

Step-by-Step Solution:

Triangles PQR and PST share angle P (given).

Angle PQR = Angle PST (given).

By AA similarity, triangle PQR ~ triangle PST.

Corresponding sides: $PQ/PS = PR/PT$.

$12/8 = 15/PT$

$12 \times PT = 8 \times 15 = 120$

$PT = 120/12 = 10$.

Why Other Options Are Wrong:

A) 8 — this is PS, not PT.

C) 12 — this is PQ, not PT.

D) 22.5 — inverts the ratio: uses $12/8 \times 15 = 22.5$ (multiplies PR by the scale factor instead of dividing).

TRAP ALERT

Option D is the inverted-ratio trap. Since PQR is the LARGER triangle ($PQ = 12 > PS = 8$), PT must be SMALLER than PR. If your answer is bigger than 15, you have the ratio backwards.

KEY INSIGHT

Always check: if the second triangle is smaller, the unknown side must be smaller than the corresponding side in the larger triangle.

FAST METHOD

Scale factor = $PS/PQ = 8/12 = 2/3$. $PT = 15 \times 2/3 = 10$.

VERDICT

Hand-solve. One proportion — 15 seconds.

Q14

Medium

Answer: A) $\sqrt{24^2 + 48^2}$

Step-by-Step Solution:

The building is 53 feet tall, and the camera is at 5 feet.

The vertical distance from camera to top = $53 - 5 = 48$ feet.

The horizontal distance is 24 feet.

By the Pythagorean theorem: distance = $\sqrt{24^2 + 48^2}$.

Why Other Options Are Wrong:

B) Uses 53 instead of 48 — forgets to subtract camera height.

C) Uses 58 — adds 5 instead of subtracting.

D) Uses 29 — adds 5 to horizontal distance instead of adjusting vertical.

TRAP ALERT

The camera is 5 feet off the ground, so the vertical leg is $53 - 5 = 48$, not 53. This "adjust for height" trap appears frequently.

KEY INSIGHT

When the observer is not at ground level, subtract the observer height from the building height to find the vertical leg.

FAST METHOD

Vertical = $53 - 5 = 48$. Distance = $\sqrt{24^2 + 48^2}$.

VERDICT

Hand-solve. Conceptual question — no real calculation needed, just setup. 15 seconds.

Q15

Medium

Answer: 15

Step-by-Step Solution:

Perimeter ratio = side ratio for similar triangles.

Scale factor = $63/42 = 3/2$.

$JK = GH \times 3/2 = 10 \times 3/2 = 15$.

TRAP ALERT

Grid-in: students may set up the ratio backwards ($42/63 = 2/3$) and compute $10 \times 2/3 = 6.67$. Check that your answer makes sense — JKL has a larger perimeter, so JK should be larger than GH.

KEY INSIGHT

The ratio of perimeters of similar triangles equals the ratio of any pair of corresponding sides.

FAST METHOD

$63/42 = 3/2$. $JK = 10 \times 3/2 = 15$.

VERDICT

Hand-solve. Direct ratio — 10 seconds.

Q16

Hard

Answer: B) $\sqrt{ab}$

Step-by-Step Solution:

In right triangle PQR with altitude PS to hypotenuse QR, three similar triangles are formed.

By the geometric mean (altitude) theorem: $PS^2 = QS \times SR = a \times b$.

$PS = \sqrt{ab}$.

Why Other Options Are Wrong:

A) $(a+b)/2$ — this is the arithmetic mean, not the geometric mean.

C) $\sqrt{a^2 + b^2}$ — this is the hypotenuse of a right triangle with legs a and b, not the altitude.

D) $ab/(a+b)$ — this is the harmonic mean divided by 2, not the geometric mean.

TRAP ALERT

Three means (arithmetic, geometric, harmonic) are common distractors. The altitude to the hypotenuse ALWAYS equals the geometric mean of the two segments.

KEY INSIGHT

Altitude to hypotenuse = $\sqrt{\text{product of the two hypotenuse segments}}$. This is a direct consequence of similar triangles formed by the altitude.

FAST METHOD

Memorize: altitude = $\sqrt{\text{seg1} \times \text{seg2}}$.

VERDICT

Hand-solve. Pure formula recall — 5 seconds if you know the theorem.

Q17

Hard

Answer: B) k^2A

Step-by-Step Solution:

For similar figures with scale factor k:

Corresponding sides scale by k.

Perimeter scales by k.

Area scales by k^2 .

Area of DEF = $k^2 \times A$.

Why Other Options Are Wrong:

A) kA — uses the linear scale factor for area (wrong dimension).

C) kP — this would be the perimeter of DEF, not the area.

D) k^2P — mixes area scaling with perimeter.

TRAP ALERT

Students confuse linear scaling (for lengths and perimeters) with quadratic scaling (for areas).

Remember: area is a 2D measure, so it scales by k^2 .

KEY INSIGHT

Lengths $\rightarrow k$, Areas $\rightarrow k^2$, Volumes $\rightarrow k^3$. This dimensional analysis principle solves many similar-figure problems instantly.

FAST METHOD

Area is 2-dimensional, so it scales by k^2 . Answer = k^2A .

VERDICT

Hand-solve. Conceptual — no computation needed. 5 seconds.

Q18

Hard

Answer: D) 20

Step-by-Step Solution:

BD $\parallel$ AE creates similar triangles BCD and ACE.

$$BC/AC = CD/CE$$

$$8/AC = 6/15$$

$$6 \times AC = 8 \times 15 = 120$$

$$AC = 120/6 = 20$$

Why Other Options Are Wrong:

A) 10 — uses $BC/CD = 8/6$ and applies to wrong segment.

B) 12 — incorrectly uses additive reasoning.

C) 18 — uses $CE - CD + BC$ or similar faulty logic.

TRAP ALERT

Make sure to identify the correct similar triangles. BD $\parallel$ AE means the small triangle BCD sits inside the larger triangle ACE.

KEY INSIGHT

When a line parallel to one side of a triangle intersects the other two sides, the smaller triangle formed is similar to the original.

FAST METHOD

$$CD/CE = BC/AC \Rightarrow 6/15 = 8/AC \Rightarrow AC = 8 \times 15/6 = 20.$$

VERDICT

Hand-solve. One proportion — 15 seconds.

Q19**Hard****Answer: C) 9/40****Step-by-Step Solution:**

$XM/MY = 3/4$ means $XM/XY = 3/7$ (since $XM + MY = XY$).

Triangle XMN is similar to triangle XYZ with scale factor $3/7$.

Area of XMN / Area of XYZ = $(3/7)^2 = 9/49$.

Area of trapezoid MNZY = Area of XYZ - Area of XMN.

Area of XMN / Area of trapezoid = $(9/49) / (1 - 9/49) = (9/49) / (40/49) = 9/40$.

Why Other Options Are Wrong:

A) $3/4$ — uses the given ratio directly without squaring.

B) $9/16$ — squares the part-to-part ratio instead of part-to-whole.

D) $9/49$ — gives the ratio of triangle XMN to the ENTIRE triangle XYZ, not the trapezoid.

TRAP ALERT

The question asks for the ratio to the TRAPEZOID (the leftover region), not to the whole triangle. D is the classic trap answer.

KEY INSIGHT

First find the ratio to the whole triangle, then subtract from 1 to find the trapezoid portion, then divide.

FAST METHOD

$XM/XY = 3/7$. Area ratio to whole = $9/49$. Trapezoid fraction = $40/49$. Ratio = $9/40$.

VERDICT

Hand-solve. Requires careful reading but the math is simple — 30 seconds.

Q20**Hard****Answer: 36****Step-by-Step Solution:**

Scale factor = $39/13 = 3$.

In the smaller triangle, hypotenuse = 13 and one leg = 5.

Other leg = $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

Longer leg of larger triangle = $12 \times 3 = 36$.

TRAP ALERT

The question asks for the LONGER leg of the LARGER triangle. Students might give 12 (longer leg of smaller) or 15 (shorter leg of larger = 5×3).

KEY INSIGHT

This is a 5-12-13 right triangle! Recognizing Pythagorean triples saves time.

FAST METHOD

Recognize 5-12-13 triple. Longer leg = 12. Scale $\times 3 = 36$.

VERDICT

Hand-solve. Pythagorean triple recognition makes this 10 seconds. Desmos adds no value.

Q21**Hard****Answer: A) $10 + 2\sqrt{5}$** **Step-by-Step Solution:**

Original triangle: A(0,0), B(10,0), C(4,8).

$AB = 10$.

$AC = \sqrt{4^2 + 8^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$.

$BC = \sqrt{(10-4)^2 + (0-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

Original perimeter = $10 + 4\sqrt{5} + 10 = 20 + 4\sqrt{5}$.

Dilation with scale factor $1/2$ multiplies every length by $1/2$.

New perimeter = $(20 + 4\sqrt{5})/2 = 10 + 2\sqrt{5}$.

Why Other Options Are Wrong:

B) $10 + 4\sqrt{5}$ — uses the original AC ($\sqrt{80} = 4\sqrt{5}$) without halving it.

C) $20 + 4\sqrt{5}$ — gives the ORIGINAL perimeter, forgetting to apply the scale factor.

D) $5 + 2\sqrt{5}$ — halves only one of the integer sides instead of both.

TRAP ALERT

Trap C gives the full original perimeter. Trap B forgets to halve the irrational side. Under dilation, ALL lengths scale by the same factor.

KEY INSIGHT

Under dilation with scale factor k , every length (including perimeter) scales by k . Compute original perimeter, then multiply by k .

FAST METHOD

Original perimeter = $20 + 4\sqrt{5}$. Divide by 2: $10 + 2\sqrt{5}$.

VERDICT

Desmos is helpful here to verify side lengths via distance formula. But the perimeter scaling step is hand-solve.

Q22**Hard****Answer: C) 128****Step-by-Step Solution:**

$TX = 5$, $XV = 3$, so $TV = TX + XV = 8$.

Scale factor of triangle TXY to triangle TVW = $TX/TV = 5/8$.

Area ratio = $(5/8)^2 = 25/64$.

Area of TXY / Area of TVW = $25/64$.

$50 / \text{Area of TVW} = 25/64$.

Area of TVW = $50 \times 64/25 = 3200/25 = 128$.

Why Other Options Are Wrong:

A) 80 — uses linear ratio $5/8$ instead of squaring it.

B) 100 — doubles the given area without using similarity.

D) 200 — uses $(8/5)$ instead of $(8/5)^2$ or makes other ratio error.

TRAP ALERT

Must SQUARE the scale factor for area. Using $5/8$ directly (without squaring) gives $50 \times 8/5 = 80$, which is trap answer A.

KEY INSIGHT

Area of similar figures scales as the square of the linear scale factor. This is tested repeatedly on the DSAT.

FAST METHOD

$(TV/UX)^2 = (8/5)^2 = 64/25$. Area = $50 \times 64/25 = 128$.

VERDICT

Hand-solve. Square the ratio and multiply — 20 seconds.

Q23

Hard

Answer: 100

Step-by-Step Solution:

Scale factor from ABC to DEF = $3 : 5$, meaning DEF is $5/3$ times larger.

Area ratio = $(\text{scale factor})^2 = (5/3)^2 = 25/9$.

Area of DEF = Area of ABC $\times 25/9 = 36 \times 25/9$.

$36/9 = 4$. Then $4 \times 25 = 100$.

Area of triangle DEF = 100.

TRAP ALERT

Grid-in trap: students might use the linear scale factor $(5/3)$ instead of squaring it. $36 \times 5/3 = 60$, which is wrong. Area scales by k -squared, not k .

KEY INSIGHT

Area scales by the SQUARE of the linear scale factor. Sides scale by k , perimeters by k , areas by k -squared. This is the most frequently tested similar-figure relationship.

FAST METHOD

Area ratio = $(5/3)^2 = 25/9$. Area = $36 \times 25/9 = 100$.

VERDICT

Hand-solve. Square the ratio and multiply — 15 seconds.

Q24

Hard

Answer: C) 60

Step-by-Step Solution:

Triangle PQR has sides 12, 16, 20 — this is a 3-4-5 right triangle (scaled by 4).

Area of PQR = $(1/2)(12)(16) = 96$.

Area of STU = 150.

Area ratio = $150/96 = 25/16$.

Scale factor = $\sqrt{25/16} = 5/4$.

Perimeter of PQR = $12 + 16 + 20 = 48$.

Perimeter of STU = $48 \times \frac{5}{4} = 60$.

Why Other Options Are Wrong:

- A) 36 — uses scale factor $\frac{3}{4}$ instead of $\frac{5}{4}$.
- B) 48 — gives the perimeter of PQR, not STU.
- D) 72 — squares the scale factor for perimeter (should only square for area).

TRAP ALERT

To go from area ratio to perimeter, take the SQUARE ROOT of the area ratio to get the linear scale factor, then multiply perimeter by that. Students who skip the square root step get answer D.

KEY INSIGHT

Area ratio $\rightarrow$ square root $\rightarrow$ linear scale factor $\rightarrow$ multiply perimeter. Recognize 12-16-20 as a 3-4-5 triple to quickly find the area.

FAST METHOD

Area of PQR = 96. Scale = $\sqrt{150/96} = \sqrt{25/16} = \frac{5}{4}$. Perimeter = $48 \times \frac{5}{4} = 60$.

VERDICT

Hand-solve. Recognizing the 3-4-5 triple speeds this up. 30 seconds.

Q25

Hard

Answer: C) 75

Step-by-Step Solution:

The surveyor creates two similar right triangles.

Small triangle: base along bank = 8 m, perpendicular = 15 m.

Large triangle: base along bank = 40 m, perpendicular = river width.

By similarity: $\frac{15}{8} = \text{width}/40$.

$8 \times \text{width} = 15 \times 40 = 600$.

$\text{width} = 600/8 = 75$ meters.

Why Other Options Are Wrong:

- A) 45 — uses wrong proportion setup.
- B) 60 — uses $40/8 = 5$, then 5×12 (wrong multiplier).
- D) 120 — multiplies $40 \times 15/8 \times 2$ (doubles the answer).

TRAP ALERT

In word problems, identifying which measurements correspond to which parts of the similar triangles is the hardest step. Draw a diagram!

KEY INSIGHT

Real-world similar triangle problems: identify the two triangles, label corresponding sides, and set up the proportion.

FAST METHOD

Scale factor = $40/8 = 5$. River width = $15 \times 5 = 75$.

VERDICT

Hand-solve. Once the diagram is clear, one multiplication — 20 seconds.

TOPIC SUMMARY

Core Concept

Similar triangles have the same shape but different sizes. All corresponding angles are equal, and all corresponding sides are in the same ratio (the scale factor). The DSAT tests this concept through proportions, parallel-line theorems, real-world applications, and area/perimeter relationships.

Key Formulas to Memorize

Corresponding sides: $a_1/a_2 = b_1/b_2 = c_1/c_2 = k$ (scale factor).

Perimeter ratio = k (same as side ratio).

Area ratio = k^2 .

Altitude to hypotenuse: $\text{altitude} = \sqrt{\text{segment1} \times \text{segment2}}$.

Most Common DSAT Question Types

1. Basic proportion setup (find a missing side) — appears in almost every test.
2. Parallel line creating similar triangles (Triangle Proportionality Theorem).
3. Shadow/height real-world problems.
4. Area ratio vs. side ratio (k^2 trap).
5. Right triangle altitude to hypotenuse (geometric mean).

Top 3 Traps to Avoid

1. Using additive reasoning instead of multiplicative (adding instead of scaling).
2. Confusing part-to-part ratio with part-to-whole ratio in parallel-line problems.
3. Using the linear scale factor for area (forgetting to square it).

Desmos vs. Hand-Solve Summary

90% of similar triangle questions are fastest by hand. Use Desmos only for coordinate geometry problems where computing multiple distances is tedious.